Consider the following procedure:

- Begin with a closed interval $I_1 = [a_1, b_1]$.
- Bisect I_1 to obtain two closed subintervals $[a_1, \frac{a_1+b_1}{2}]$ and $[\frac{a_1+b_1}{2}, b_1]$.
- Select one of the two subintervals above, and call it $I_2 = [a_2, b_2]$.
- Keep repeating this process to obtain a sequence of intervals $I_1, I_2, I_3, I_4, I_5, \ldots$.

1. Show that the sequence of intervals $I_1, I_2, I_3, I_4, I_5, \ldots$ obtained above satisfies both hypotheses of the Nested Intervals Theorem. That is, show that

   (i) $a_n \leq a_{n+1}$ for all $n \in \mathbb{N}$ and $b_{n+1} \leq b_n$ for all $n \in \mathbb{N}$, so that $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$, and

   (ii) $\inf\{b_n - a_n\} = 0$.

2. By the result of Question 1, the Nested Intervals Theorem implies $\bigcap_{n=1}^{\infty} I_n = \{x\}$ where $x = \sup\{a_n\} = \inf\{b_n\}$. Show that any open interval containing x contains I_N for some $N \in \mathbb{N}$.

Example

We now give an example of using the bisection procedure above as a method of proof, which we call the *Bisection Method*.

Rationals between reals, revisited.

Use the Bisection Method to give an alternative proof that, given any $x \in \mathbb{R}$ and $\varepsilon > 0$, the open interval $(x-\varepsilon, x+\varepsilon)$ contains a rational number. The case $x \in \mathbb{Q}$ is easy. (Why?) To help you get started with the case $x \notin \mathbb{Q}$, begin by showing that there exists a closed interval $I_1 = [a_1, b_1]$ containing x where both a_1 and b_1 are rational. Then explain, relative to the given x, how to choose each subinterval I_n and explain why the endpoints of each I_n are rational. The rest of the proof is then a straightforward application of the Nested Intervals Theorem.